Outcome Regression & Propensity Scores

1. Business Use Case & Problem Formulation

In telecommunications customer retention, offering targeted discounts or loyalty perks (treatment A = 1 vs A = 0) is a primary operational lever to reduce 90-day customer churn (outcome Y, where 1 = churned, 0 = retained). However, in historical observational data, retention offers are non-randomly targeted toward subscribers with high contract age, frequent support tickets, or high monthly charges (covariates L). Simple naive comparisons between offered and non-offered subscribers produce severe confounding bias.

Variable Name	Role in Causal Model	Business Metric / Operational Definition
Retention Offer (A)	Binary Treatment	1 = Offered 20% Bill Discount for 6 Months, 0 = Standard Billing
Customer Churn (Y)	Binary Outcome	1 = Account Terminated within 90 Days of Offer, 0 = Active
Customer History (L)	Confounder Vector	Tenure, Monthly Charges, Support Tickets, Data Usage, Auto-Pay

2. Outcome Regression Modeling

Outcome regression directly models the conditional expectation of the outcome E[Y | A, L] as a function of treatment A and confounders L. For a binary churn outcome, we fit a parametric logistic regression model:

$$\text{logit } P(Y = 1 \mid A, L) = \alpha_0 + \alpha_1 A + \alpha_2 (A \times L) + \alpha_3 L$$

Under conditional exchangeability ($Y^a \perp A \mid L$), positivity, and correct model specification, the regression parameters allow us to compute stratum-specific conditional causal effects or standardize over L to obtain the Average Treatment Effect (ATE).

> **Fine Point 15.1: Nuisance Parameters in Outcome Regression**
> Parameters α_0 and α_3 in outcome regression are nuisance parameters. They model the baseline risk $E[Y^{a=0} \mid L]$ rather than the causal contrast itself. If the baseline risk model is misspecified, outcome regression estimates of the treatment effect α_1 become biased. In contrast, semiparametric methods like G-estimation are agnostic about baseline risk.

3. Propensity Scores & The Balancing Property

The **propensity score**, denoted $\pi(L) = P(A = 1 \mid L)$, is the conditional probability of receiving treatment given measured confounders L. Rosenbaum & Rubin (1983) proved a fundamental theorem: conditional exchangeability given L implies conditional exchangeability given the scalar propensity score π(L):

$$Y^a \perp A \mid L \Rightarrow Y^a \perp A \mid \pi(L)$$

The propensity score acts as a **balancing score**: conditional on π(L), the distribution of all measured confounders L is identical between treated (A=1) and untreated (A=0) subscribers ($A \perp L \mid \pi(L)$). For an individual unit n, the Inverse Probability Weight (IPW) is defined as:

$$W_n = A_n / \pi_n(L_n) + (1 - A_n) / (1 - \pi_n(L_n))$$

4. Propensity Score Implementation Strategies

Method	Operational Procedure	Key Advantages	Primary Limitations
Propensity Stratification	Divide subscribers into 5-10 deciles of π(L). Estimate ATE within deciles and average.	Non-parametric regarding outcome relationship.	Residual confounding within wide deciles; arbitrary boundaries.
Outcome Regression on π(L)	Fit outcome model E[Y | A, π(L)] using splines or polynomial terms.	Smooth adjustment; easily incorporates treatment interaction terms.	Requires correct functional form specification for π(L).
Propensity Score Matching (PSM)	Pair each treated subscriber (A=1) with untreated subscribers having close π(L).	Creates intuitive balanced matched cohort; restricts to common support.	Excludes unmatched units; changes target population to ATT.
Inverse Probability Weighting (IPW)	Weight each unit by W_n to construct a pseudo-population.	Retains full sample size; estimates marginal structural models.	Unstable when π(L) is near 0 or 1; high variance without stabilization.

5. Causal Models vs. Predictive Machine Learning Models

A widespread, dangerous misconception in industrial analytics is treating propensity models as standard predictive ML classifiers. **Predictive power (e.g., high AUC or C-statistic) is NOT the goal of a propensity model.** The sole objective is to block all backdoor paths and satisfy conditional exchangeability.

> **Warning: Pitfalls of Predictive Variable Selection in Propensity Models**
> **1. Including Colliders or Intermediate Variables:** Induces severe selection bias (M-bias).
> **2. Including Pure Treatment Predictors (Instruments):** Including variables Z that cause treatment A but have no direct relation to outcome Y severely inflates estimator variance and amplifies unmeasured confounding bias.
> **3. Maximizing C-statistic / AUC:** A propensity score model that perfectly predicts treatment (AUC ≈ 1.0) destroys positivity (no overlap), rendering causal estimation impossible!

6. Doubly Robust Plug-in Estimators (TMLE / Bang & Robins)

To protect against model misspecification, **Doubly Robust (DR) Estimators** combine an outcome regression model m(A,L) = E[Y | A,L] and a propensity score model π(L) = P(A=1 | L). A DR estimator is consistent if **EITHER** the outcome model OR the propensity model is correctly specified.

Implementation via Bang & Robins Clever Covariate Method:
1. Estimate IP weights $W_n = A_n / \pi(L_n) + (1 - A_n) / (1 - \pi(L_n))$.
2. Define the 'clever covariate' $R = W_n$ for A=1 and $R = -W_n$ for A=0.
3. Fit an outcome logistic regression model adding R as an explicit covariate: **logit P(Y=1 | A, L, R) = $\beta_0 + \beta_1 A + \beta_2 L + \theta R$**.
4. Predict counterfactual churn risks $\hat{Y}^{a=1}$ and $\hat{Y}^{a=0}$ for all subscribers under A=1 and A=0, then average across the population to get the unbiased Average Treatment Effect.

7. Methodological Comparison & Recommendations

Criterion / Property	Outcome Regression	Propensity Stratification / Matching	Doubly Robust (DR / TMLE)
Model Requirement	Correct model for E[Y | A,L]	Correct model for P(A=1 | L)	Correct model for EITHER E[Y | A,L] OR P(A=1 | L)
High-Dimensional L	Prone to outcome model misspecification	Collapses L to 1D scalar π(L)	Optimal: uses ML with cross-fitting
Positivity Sensitivity	Extrapolates over empty cells (hidden bias)	Explicitly exposes lack of overlap / support	Robust; bounded clever covariates prevent extreme weights
Target Population	Full target population (ATE)	Often restricted to matched subset (ATT)	Full target population (ATE)

Production Telco Pipeline Recommendations

• Naive outcome comparisons overestimate retargeting efficacy (-9.2% churn risk) due to severe confounding.

• Correctly specified parametric methods (Outcome Regression, Propensity Stratification, PSM) converge on a true causal ATE of **-4.5% to -4.8%**.

• **Doubly Robust estimation (TMLE)** is the recommended standard for production retention pipelines, providing dual protection against model misspecification.